Colour. Section 2.

The duality of the concept of color:

- Color as a physical property of reality (the physical nature of color)
- Color as a subjective psychophysiological sensation that translates into certain emotional states that differ from person to person (and this ambiguity is of main interest to fine art)

The physical nature of color

1. General situation

According to the electromagnetic theory of light (Maxwell’s theory), light is electromagnetic vibrations, i.e. periodic changes in electric and magnetic fields in space. According to the same theory, different types of radiation energy have the same nature and differ only in wavelength.

The human eye is able to perceive color only at a wavelength of 400 to 700 nanometers. This is a very narrow range of values relative to the entire range of electromagnetic radiation lengths.

Each wavelength in the visible range corresponds to a specific color (as we see it on the Newton’s circle), with violet rays having the shortest length and red rays having the longest.

Color	**Wave lenght**	**Oscillation frequency frequency (bilion in seconds)**
Red	800 – 650	400 – 470
Orange	640 – 590	470 – 520
Yellow	580 – 550	520 – 590
Green	530 – 490	590 – 650
Light blue	480 – 460	650 – 700
Blue	450 – 440	700 – 760
Violet	430 – 390	760 – 800

2. Spectral composition of light

Sunlight contains all the wavelengths of the visible spectrum, which cannot be said about all artificial lighting devices, which can produce a luminous flux consisting only of a part of the range of visible spectrum waves.

3. Light temperature

The light temperature is measured in Kelvin. 2500 warm light, 3000 incandescent lamp "yellow", 6400, if I'm not mistaken, daylight in cloudy weather. It's like a shade of light that results from an uneven spectral composition.

As mentioned above, each light source provides a certain part of the visible spectrum, or the entire visible spectrum. And here, if the waves predominate in this light, for example, long ones, then the light is warmer, if short, then it is colder.

That is, the temperature of the light can tell us about the distribution of the spectral composition, but we must take into account that it does not tell us anything about the completeness of the spectrum of light from the source in relation to the entire range of visible waves.

Goethe's Color Harmony

Goethe's "The Doctrine of Color" contains a color model different from Newton's circle. Goethe's color wheel contains, in addition to "additional" (harmonious/complementary, oppositional to each other) color combinations, "characteristic" (in diameter and large chord), "characterless" and "weak" (when colors of the same color tone, but of different saturation, are compared). Goethe's experiments with colored shadows showed that diametrically opposed (complementary) colors are precisely those that mutually evoke each other in the viewer's mind. [1]

A harmonious color, according to Goethe, arises "when all neighboring colors are brought into balance with each other." But harmony, according to Goethe, despite all its perfection, should not be the ultimate goal of an artist, because harmony always has "something universal and complete, and in this sense devoid of character." Let's recall Arnheim's model of perception.

It is possible to build the color of a work based on the method of shifting all the colors to any one color, as if we were looking at it through colored glass. This technique is often used in both painting and cinematography, but it is far from the most perfect way to achieve color harmony. There are more advanced and elegant methods that, however, require more work and a higher visual culture. And this method does not create integrity, but only uniformity.

Munsell Color Body

The Munsell color body [2] is a color body in a 3-dimensional space, the dimensions of which are called TONE, SATURATION, LIGHTNESS (Hue, Luminance, Saturation). Munsell's idea was that

the maximum saturation of different tones is achieved at different levels of lightness (as an example, we can cite the fact that saturated yellow will always be lighter than saturated purple) and at the same time the length of the vector of luminous gradations for each tone is not the same. Based on this, the Munsell color body has a complex asymmetrical shape. Each slice of this body contains a plane in terms of LIGHTNESS, which displays colors in terms of TONE and SATURATION in a single measure of LIGHTNESS.

[note] The Munsell color body is much broader in scope than the HSL model.

Above, in Goethe’s color harmony, color combinations are mentioned: "additional", "characteristic", "characterless/weak". These combinations take into account the pure tone of the color, but do not take into account the lightness. This is unacceptable if we are talking about real harmony in the visual arts, where this value is always taken into account when comparing colors. Therefore, the true lines of harmonization in the color space are not only in the plane of the color circles, but pass from one circle to another as secants in all directions of the color body.

"Changing the lightness of the combined colors is one of the most powerful and universal means of harmonization. In practice, the exact color similarity in a color (for example, a suit) is not as important as the fact that it is well lit or in shadow, or overexposed. By changing the contrast of lighting on an object, you can unrecognizably change the entire color harmony in the frame, while some colors will go out (although under other lighting conditions they would look quite bright and saturated), and others will flare up as bright color accents (although under other lighting conditions these colors do not visually or psychologically stand out.)"

"Color and contrast. Technology and creative choice" by V.Zheleznyakov

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SUBJECT and APERTURE color

Object color is the color of the surfaces of objects that we see under certain lighting conditions.

Aperture color is the color of a uniform light field visible to us from a source, lamp, sun, cloudy sky, or fire.

Our artistic task, as I understand it, is to see its true color in these lighting conditions when working with nature, and to try to minimize the influence of the mechanism of constancy on our perception. Or to put it another way, to perceive the object color as if it were an aperture, that is, to see the color of the light reflected by the object itself, rather than perceiving

color as an attribute of the object / a constant property of the surface.

"We can say that objective color is not so much reality as a perceptual representation of reality, because it depends on so many reasons and is a conditional value.<...> In fact, the quality of a color-reproducing system is not determined by the dye yield or the amount of color saturation. The main thing is the ability to reproduce as accurately as possible all the color changes during the transition from one light level to another as a result of lightening to almost white and darkening to almost black. This is what is called valera in painting and is of fundamental importance for the entire practice of fine art."

"Color and contrast. Technology and creative choice" by V.Zheleznyakov

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Literature:

- "Zheleznyakov V.N. Color and contrast. Technology and creative choice. Moscow: VGIK, 2001"
- "Sergey Mikhailovich Lando. Photo composition for film school"

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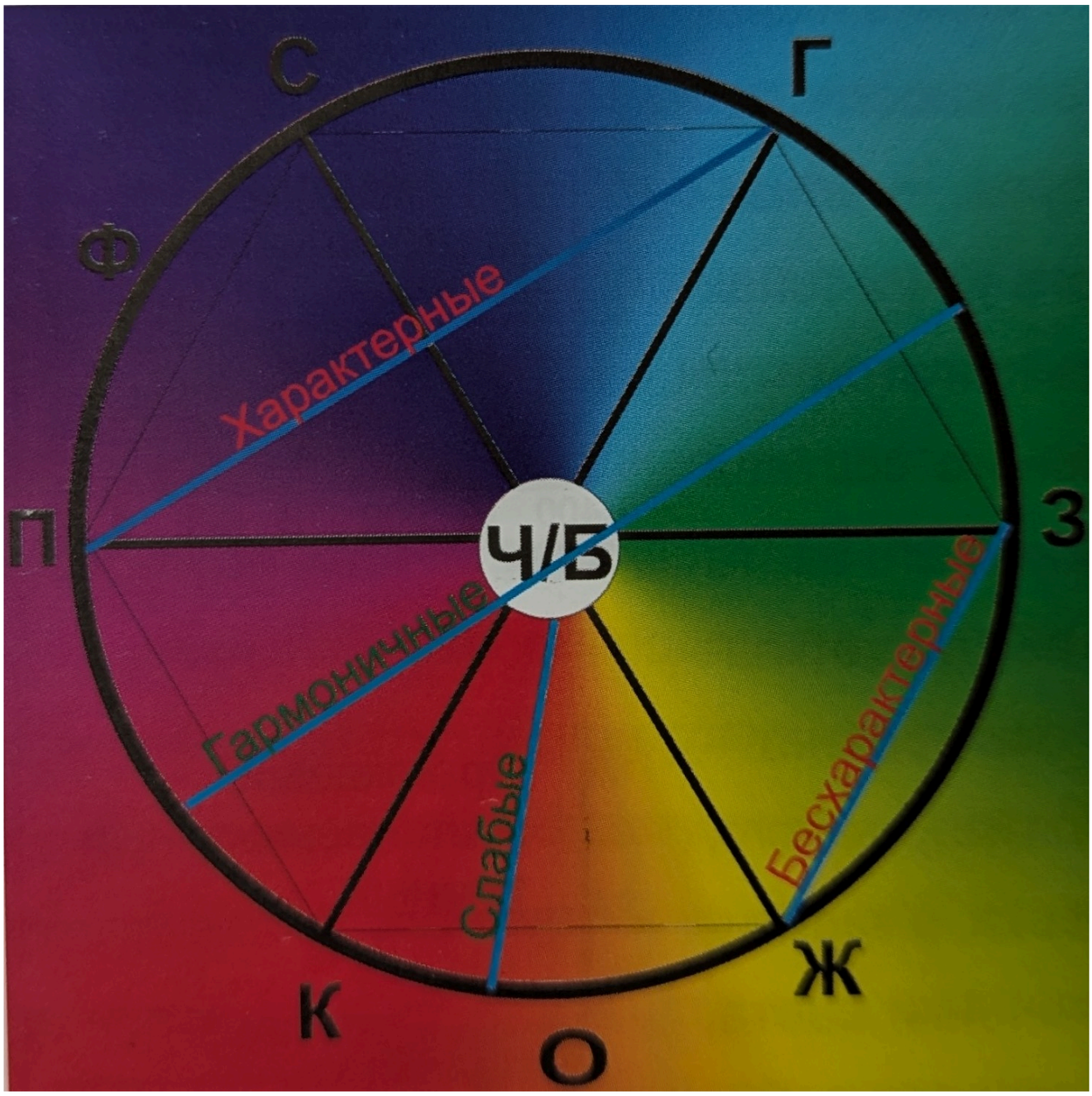
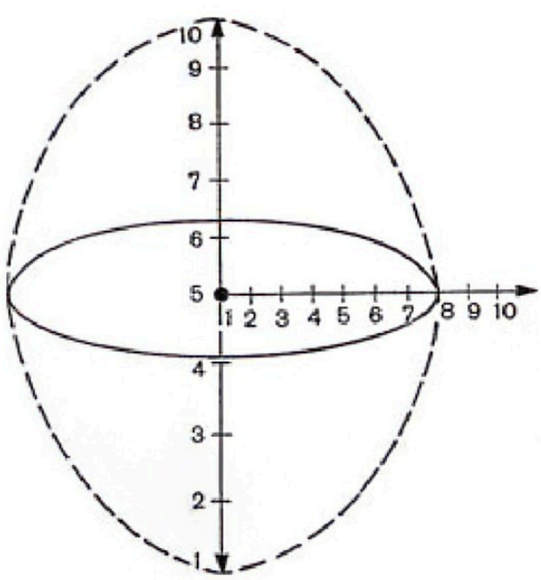
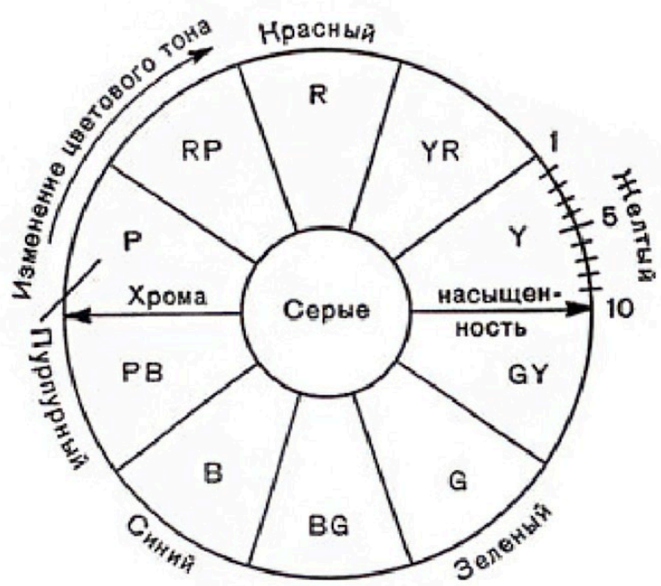
[1]this image was borrowed from Sergei Mikhailovich Lando’s book "Photographic Composition for Film School"



[2]



Илл.15 Цветовое тело Манселла.